PHY 112L
A 2-D Ideal-Gas Atmosphere

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1 Introduction

In this assignment, we will model an atmosphere as an ideal gas in two dimensions. We will measure the heat exchanged during the approach to thermal equilibrium, numerically determine the heat capacity of an ideal gas, and compare with our theoretical expectation. We will use the heat capacity to understand adiabatic compression, and simulate it. Next, we will add gravity to our model and measure how this impacts the molecular density and heat capacity of the atmosphere.

2 Heat Capacity of an Ideal Gas

For an ideal gas (without gravity) the total energy U is purely kinetic, and the equipartition theorem implies that:

$$U = \frac{f}{2}NkT \tag{1}$$

where f is the number of quadratic degrees of freedom: 3 for an ideal gas in 3-D, and 2 for an ideal gas in 2-D.

The heat capacity C describes quantitatively how the temperature of the gas changes as heat is added or removed:

$$C \equiv \frac{Q}{\Delta T} \tag{2}$$

Under constant volume, no work is done on the gas, and so:

$$\Delta U = Q + W = Q$$

Therefore, we can calculate the heat capacity under constant volume as:

$$C_V \equiv \left(\frac{Q}{\Delta T}\right)_V = \left(\frac{\Delta U}{\Delta T}\right)_V$$

and taking ΔT small, we have:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V \tag{3}$$

Applying this to an ideal gas we find that:

$$C_V = \frac{f}{2}Nk \tag{4}$$

When two systems, A and B, initially at temperatures T_A and T_B are brought into thermal contact, they reach an equilibrium temperature T by exchanging heat. The conservation of energy implies that:

$$C_A T_A + C_B T_B = C_A T + C_B T$$

where C_A and C_B are the heat capacities of the two systems. Equivalently:

$$T = \frac{C_A}{C_A + C_B} \cdot T_A + \frac{C_B}{C_A + C_B} \cdot T_B \tag{5}$$

Notice that for an infinite reservoir R with $C_R = \infty$ this reduces to:

$$T = T_R$$

Also, in the case that $C_A = C_B$, we have:

$$T = \frac{T_A + T_B}{2}$$

that is, the temperature is just the average of the two systems.

3 Adiabatic Compression and the Heat Capacity

In HW 1 we derived the pressure and work for adiabatic compression of an ideal gas, directly from $U = \frac{f}{2} NkT$. Here we derive them again from the heat capacity. During an adiabatic process no heat enters or leaves the gas, $Q = 0$, so each small change in the energy of the gas is work done on it. For our 2-D gas, with pressure O and area A:

$$dU = dW = -O\, dA$$

At constant area, the same change in temperature dT would require a heat $C_V\, dT$, and because the energy of an ideal gas depends only on its temperature, $dU = C_V\, dT$ in any process. So:

$$C_V\, dT = -O\, dA = -\frac{NkT}{A}\, dA$$

using the ideal gas law $OA = NkT$. Dividing by $C_V T$:

$$\frac{dT}{T} = -\frac{Nk}{C_V} \frac{dA}{A}$$

and integrating both sides:

$$T\, A^{Nk/C_V} = \text{const} \tag{6}$$

For our 2-D ideal gas, $C_V = Nk$ (Equation 4 with $f = 2$), so during adiabatic compression:

$$T\, A = T_1 A_1 \tag{7}$$

and with $O = NkT/A$:

$$OA^2 = O_1 A_1^2$$

as we found in HW 1. The work done on the gas compressing it from A_1 to A_2 is simply the change in its energy:

$$W = \Delta U = C_V\, (T_2 - T_1) = NkT_1 \left(\frac{A_1}{A_2} - 1 \right) \tag{8}$$

which agrees with the integral $-\int O\,dA$ from HW 1. For the gas cylinder of HW 1, compressed from height h_1 to h_2, $A_1/A_2 = h_1/h_2$, and in our system of units:

$$\frac{O}{O_0} = \left(\frac{T_1}{T_0}\right)\left(\frac{ah_1}{h^2}\right) \tag{9}$$

$$\frac{W}{U_0} = \frac{T_1}{T_0}\left(\frac{h_1}{h_2} - 1\right) \tag{10}$$

For example, compressing the gas from $h_1 = 4a$ to $h_2 = 2a$ starting at $T_1 = T_0$ doubles its temperature, to $T_2 = 2T_0$, and takes work $W = 1\,U_0$.

4 Ideal Gas with Gravity

Suppose now that our ideal gas is in a constant gravitational field with acceleration g. Then each particle has gravitational potential energy:

$$E_{\text{grav}} = mgy \tag{11}$$

where we have set $y = 0$ as the zero potential. We expect the position of the particles to follow a Boltzmann distribution, so that:

$$p(y) \propto \exp\left(-\frac{E_{\text{grav}}(y)}{kT}\right) = \exp\left(-\frac{mgy}{kT}\right)$$

The normalization condition:

$$\int_0^\infty p(y)dy = 1$$

gives us the normalization constant, and so:

$$p(y) = \frac{mg}{kT}\exp\left(-\frac{mgy}{kT}\right) \tag{12}$$

This is a decaying exponential, so that the atmosphere gets thinner as you go higher, exactly as we expect. The average height of the gas molecules is:

$$\langle y\rangle = \frac{kT}{mg} \tag{13}$$

The total gravitational potential energy for the entire gas is:

$$U_{\text{grav}} = \sum_i mgy_i = Nmg\,\langle y\rangle$$

and using Equation 13 we have:

$$U_{\text{grav}} = NkT$$

The total energy U of the system includes the kinetic energy part, so:

$$U = \frac{f}{2}NkT + NkT = \frac{f+2}{2}NkT \tag{14}$$

thus the heat capacity of our atmosphere is:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{f+2}{2}Nk \tag{15}$$

Notice the system is still at constant V (just V is now infinite) or, more importantly, no external work is done on the atmosphere, so our previous result for the kinetic part still applies. The influence of gravity increases the heat capacity by Nk. This additional heat capacity comes from the system's ability to stash additional energy as gravitational potential energy. This situation is quite analogous to a gas under constant pressure, for which the same additional term in the heat capacity is obtained.

5 System of Units

We will use the same system of units as in the previous assignment with one additional constraint. Interestingly, the ideal gas model of HW 1 had no specific length scale defined: we simply took it to be some scale of interest which we called a. Our results apply equally well to the air in a child's balloon or the vast extremely low-density plasmas of the interstellar medium. Once we turn on gravity, however, there is a relevant length scale given by Equation 13. So we set:

$$a \equiv \frac{kT_0}{mg} \tag{16}$$

so that:

$$\langle y/a \rangle = \frac{T}{T_0} \tag{17}$$

There is no need to change any of our results from the previous homework. **The quantity a will still be set to 1 in your code.** It's only in interpreting our results that we would be forced to apply Equation 16.

The other units that we defined in the previous homework continue to apply, so let's recall them here. Motivated by the Maxwell-Boltzmann distribution, the velocity unit v_0 is defined by:

$$v_0^2 \equiv \frac{kT_0}{m}$$

and the unit for time is:

$$\tau \equiv a/v_0$$

Notice that our units of acceleration are now:

$$\frac{a}{\tau^2} \equiv g$$

Motivated by the ideal gas law, we can set the unit of pressure to:

$$O_0 \equiv \frac{NkT_0}{a^2}$$

and the unit of total energy:

$$U_0 = NkT_0$$

so that the unit of heat capacity is $U_0/T_0 = Nk$. Remember that for analytic work, we keep these factors in place. But when using an analytic result within our computer programs, we set all of these quantities to 1:

$$m = a = g = T_0 = v_0 = \tau = O_0 = U_0 = 1$$

However, if you would like, you can turn off gravity in a specific context by setting $g = 0$. Just take care that you turn gravity on with $g = 1$. As before, the temperature of an ideal gas is still calculated through the average kinetic energy, which in our system of units becomes:

$$\frac{T}{T_0} = \frac{\langle v_x^2/v_0^2 \rangle + \langle v_y^2/v_0^2 \rangle}{2} \tag{18}$$

This is also the total kinetic energy of the gas, in units of total energy. The energy of any group of molecules, e.g. the heat Q given up by reservoir particles, is measured in the same unit: a sum of $\frac{1}{2}(v_x^2 + v_y^2)/v_0^2$ over the molecules, divided by N. The total energy of the gas has a new term from the gravitational potential energy, so that:

$$\frac{U}{U_0} = \frac{\langle v_x^2/v_0^2\rangle + \langle v_y^2/v_0^2\rangle}{2} + \langle y/a\rangle \tag{19}$$

Using Equations 17 and 18 we can write this rather elegantly as:

$$\frac{U}{U_0} = 2 \cdot \frac{T}{T_0} \tag{20}$$

Recalling that:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

we see that in our units, the heat capacity of our gas under the influence of gravity is:

$$\frac{C_V}{U_0/T_0} = 2 \tag{21}$$

If we turn off gravity, we lose the gravitational potential term in the total energy and have:

$$\frac{U}{U_0} = \frac{T}{T_0} \tag{22}$$

and:

$$\frac{C_V}{U_0/T_0} = 1 \tag{23}$$

Notice that you can also get these results for C_V from Equations 4 and 15 if you simply divide by

$$U_0/T_0 = Nk$$

and set $f = 2$.

6 Bouncing under Gravity

When simulating our atmosphere under the influence of gravity, we'll have to adjust our algorithm for circulating the gas and bouncing off the ground. We'll no longer need a top for our cylinder of gas: we'll allow particles to reach as high as they can under the influence of gravity.

Our numerical model includes the y-position of each molecule. While inside the container, a molecule's position can be updated for a time step Δt as:

$$y \to y + v_y\, \Delta t - \frac{1}{2}\, g\, (\Delta t)^2 \tag{24}$$

and the velocity by:

$$v_y \to v_y\ - g\Delta t \tag{25}$$

There is no longer any need to bounce at the top of the cylinder, as the last term will inevitably pull molecules back toward $y = 0$. We will, however, have to handle particles bouncing off the bottom of the container at $y = 0$, as illustrated in Fig. 1. Suppose that after a time step Δt, we

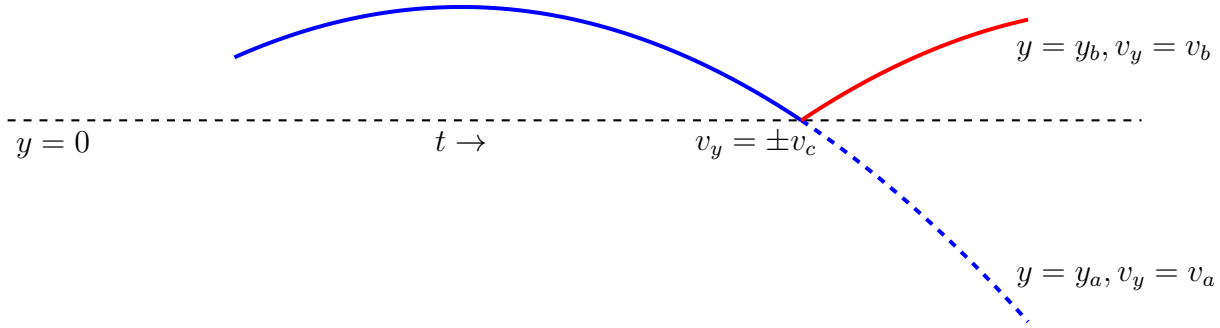


Figure 1: After a timestep Δt shown in blue, some molecule trajectories pass through $y = 0$, as if the ground was not there, ending at $y = y_a < 0$ and $v_y = v_a < 0$. The bouncing algorithm finds y_b and v_b, following the bounced red trajectory instead, and ending at the same time.

have a particle that passed through the bottom of the container, with $v_y = v_a$ and $y = y_a$. We will certainly have:

$$y_a < 0$$

and

$$v_a < 0.$$

Bouncing is not so simple in this case, due to the acceleration due to gravity. Our approach will be to find position y_b and velocity v_b if the particle had bounced, instead of simply passing through $y = 0$. Our first step is to find the time t since we passed through $y = 0$, with $y = 0$ and $v_y = v_c$, the velocity at the crossing. (We call it v_c to keep it apart from our unit of velocity v_0.) We do not yet know either v_c or t, but they are easy to calculate from 1-D kinematics:

$$v_a = v_c - gt$$

so that:

$$t = \frac{v_c - v_a}{g}$$

and also:

$$v_a^2 = v_c^2 - 2gy_a$$

so:

$$v_c = -\sqrt{v_a^2 + 2gy_a}$$

notice that we have kept the negative square root corresponding to the most recent crossing of $y = 0$ with $v_c < 0$. (The other root corresponds to an earlier time with the molecule passing upwards through $y = 0$.) And so:

$$t = \frac{-v_a - \sqrt{v_a^2 + 2gy_a}}{g}$$

That result for t is completely valid, except it is numerically unstable. In the vicinity of $y_a = 0$, which is where we are located, the $2gy_a$ will be overcome by numerical roundoff errors from calculating:

$$-v_a - \sqrt{v_a^2} = |v_a| - |v_a| = 0$$

Instead we use the algebraically equivalent but numerically stable form:

$$t = \frac{-2y_a}{\sqrt{v_a^2 + 2gy_a} - v_a} \tag{26}$$

which we obtain by multiplying the top and bottom by the conjugate.

With v_c and t now in hand, we can calculate the position y_b and the y-component of velocity v_b had the particle bounced at $y = 0$, which is to say $v_c \to -v_c$. So:

$$y_b = (-v_c t) - \frac{1}{2}\, g t^2 \tag{27}$$

and:

$$v_b = (-v_c) - g\, t \tag{28}$$

Take care that $-v_c$ is positive! Also notice that if t is large enough, the particle may have fallen below $y = 0$ again, so that $y_b < 0$. You will have to keep bouncing molecules until all of them have $y \geq 0$ at the end of the time step.

7 Required Functions

Define these functions at the top level of your notebook (not inside another function), and use them in your solution. **Please stick to these function names and definitions.** It will help us help you debug your code.

- From HW 1, unchanged: `collide(ax, ay, bx, by, theta)`, `temp(vx, vy)` and `move(y, vy, h, vp, dt)`.
- Thermal contact with a reservoir at temperature `Tr`, for `ncoll` collisions:

```
def collide_reservoir(vx, vy, Tr, ncoll):
    # your code here
    return vx, vy, Q # updated values, and the heat given to the gas
```

  Each collision uses a new reservoir particle, drawn as in HW 1, and a randomly chosen gas molecule. `Q` is the kinetic energy the reservoir particles lose in the collisions, in units of U_0 (see the System of Units).

- Thermal contact between two gases A and B, for `ncoll` collisions:

```
def collide_gases(vxa, vya, vxb, vyb, ncoll):
    # your code here
    return vxa, vya, vxb, vyb # updated values
```

  Each collision is between a randomly chosen molecule of gas A and a randomly chosen molecule of gas B.

- The motion of the molecules under gravity, including all bouncing at $y = 0$:

```
def move_gravity(y, vy, g, dt):
    # your code here
    return y, vy # updated values
```

  Keep bouncing until every molecule has $y \geq 0$. Test values: a single molecule at rest at $y = 1$ falls and bounces once in $\Delta t = 2$, and twice in $\Delta t = 5$:

```
print(np.around(move_gravity(np.array([1.0]), np.array([0.0]), 1, 2), 3))
print(np.around(move_gravity(np.array([1.0]), np.array([0.0]), 1, 5), 3))
```

  should output:

```
[[0.657]
 [0.828]]
[[0.784]
 [0.657]]
```

8 Checking Your Notebook: hw2check

As in HW 1, this assignment comes with a checker, `hw2check.py`, available with the assignment on the course website. Put it in the same folder as your notebook. If you work on a cloud server (e.g. JupyterHub or Google Colab), upload `hw2check.py` next to your notebook there. Do not modify it. Use it as follows:

- **Import it** in the first cell of your notebook, along with everything else you need:

```
# Setup
import numpy as np
import matplotlib.pyplot as plt
import hw2check
```

- **Label your cells.** The first line of every other code cell is a comment naming the problem it belongs to: `# Problem 1`, `# Problem 2`, ..., `# Problem 6`.

- **Required functions.** Define the functions listed above, exactly as described.

- **Report each result** once, with `hw2check.report`, in the cell where you calculate it:

```
hw2check.report("P1 C", C)
```

  The results to report are listed with each problem.

- **Register each required plot** by calling `hw2check.figure` right after making it, in place of `plt.show()`:

```
hw2check.figure("P1b")
```

  The required plots are `"P1a"`, `"P1b"`, `"P3"` and `"P5"`.

- **Run the checker in the last cell:**

```
import hw2check; hw2check.run()
```

  It tests your functions with the test values in this assignment, checks that every result and plot is there, tells you whether each result is correct to within the expected precision, and warns you about signs of cells run out of order. It ends either with "ready to submit" or with a list of what is still to do.

Before submitting, use Kernel → Restart & Run All, and check that the last cell says "ready to submit". If you cannot get there, add a text (markdown) cell at the end of your notebook explaining why. An honest result outside the expected precision, with your explanation, is better than one tuned to pass. Passing the checker does not guarantee credit: we run additional tests when grading.

9 Homework Problems

Please submit one PDF file (including output) and one python notebook file (.ipynb) of your completed assignment, after Restart & Run All, with `hw2check.run()` in the last cell. This assignment must be done in Python, in a Jupyter notebook. You will receive **zero credit if you do not use our system of units consistently!** When asked to calculate a quantity, e.g. temperature, answer in our units, e.g. T_0. Do not first calculate in SI units and then convert!

The starting point for this assignment is Problem 4 from the previous homework. You will want to initialize a numerical model for an ideal gas tracking variables v_x, v_y, and y for $N = 10000$ molecules, with height $H = 4a$. Use `collide_reservoir` to bring your gas into equilibrium with a reservoir at $T_R = 1\,T_0$.

Problem 1: Start with the gas at $T = 1\,T_0$ and bring it into thermal contact with a reservoir at $T_R = 2\,T_0$ in ten steps: call `collide_reservoir` ten times, each time for $n_{\text{coll}} = N = 10000$ collisions, and record the gas temperature and the heat exchanged after each call. Plot Q versus iteration (from 1 to 10) and T versus iteration (from 1 to 10). You should see the Q values are large for the first few iterations and then oscillate near zero, and the temperature rising steadily until it reaches $T = 2\,T_0$. This should convince you that it takes about ten times as many collisions as gas molecules, $n_{\text{coll}} = 10N = 100,000$, to reliably bring your gas into thermal equilibrium. Register this plot as `"P1a"`.

Next, bring your gas into thermal contact with a reservoir at $T_R = 3\,T_0$, with a single call for $10N$ collisions, and record the temperature and the heat exchanged. Repeat with reservoirs at $4\,T_0$ and $5\,T_0$. Plot the total heat exchanged as a function of the temperature at each of the four temperature points ($2\,T_0$ to $5\,T_0$). For example, the total heat exchanged at temperature $T = 3\,T_0$ is the sum of all the Q exchanged with the reservoir at $T_R = 2\,T_0$ and the Q exchanged with the reservoir at $T_R = 3\,T_0$. Read off the heat capacity of your gas from the plot, and compare it to your expectation in the comments. Report the four temperatures as an array `"P1 T"`, the four totals of heat as `"P1 Q"`, and your heat capacity as `"P1 C"`. Register this plot as `"P1b"`.

Problem 2: Call your current gas model "gas A". Create a second gas model "gas B" with the same parameters. Initialize gas A to $T = 1\,T_0$ and gas B to $T = 2\,T_0$. Bring gas A and gas B into thermal contact with each other, using `collide_gases`, for $n_{\text{coll}} = 10N = 100,000$ collisions. Measure the temperature of both after the collisions and compare, in the comments, with the prediction of Equation 5: the two gases have the same heat capacity, $C_A = C_B = 1\,U_0/T_0$, so both should reach the average, $T = 1.5\,T_0$. Report them as `"P2 TA"` and `"P2 TB"`.

Problem 3: Simulate adiabatic compression of gas A, starting at $T = 1\,T_0$, from $h = 4a$ to $h = 2a$ in 80 timesteps of size $\Delta t = 10\tau$, with your `move` function from HW 1. Set the speed of the piston appropriately, and calculate the pressure at each step as in HW 1. The collisions with the piston bring the gas out of thermal equilibrium at each step. This is an adiabatic process, so you cannot use a reservoir. Instead, bring the gas back into thermal equilibrium after each step by colliding $N = 10000$ randomly chosen pairs of its own molecules, as in Problem 3 of HW 1. Plot your measured O versus A and compare to your expectation from Equation 9. Integrate your curve to get the work done, and compare to your expectation from Equation 10. Measure the final temperature, and check that $W = C_V\,\Delta T$ (Equation 8). Report the area $A = ah$ and the pressure O of each step as arrays `"P3 A"` and `"P3 O"`, the work as `"P3 W"`, and the final temperature as `"P3 T"`. Register

your plot as `"P3"`.

Problem 4: Use your `move_gravity` function to turn gas A into a model for the atmosphere with gravity and $T = 1\,T_0$. To do so use[1] a total of $n_{\text{coll}} = 20N = 200,000$ collisions and $n_{\text{step}} = 2000$ time steps of size $\Delta t = 0.01\tau$. During each time step, account for gravity and then simulate thermal contact with a reservoir at $T_R = 1\,T_0$ for $n_{\text{coll}}/n_{\text{step}}$ collisions. Measure the temperature of the gas and the total energy (kinetic and gravitational) and compare with your expectation. Report them as `"P4 T"` and `"P4 U"`.

Problem 5: Make a histogram of the y-position of the molecules, and compare to the theoretical prediction of Equation 12. Report the mean height $\langle y \rangle$ as `"P5 y"`, and register your plot as `"P5"`.

Problem 6: Bring your atmosphere (gas A) at temperature $T = 1\,T_0$ into thermal contact with gas B at temperature $T = 4\,T_0$. Apply gravity to gas A but not to gas B. Use a total of $n_{\text{coll}} = 20N = 200,000$ collisions, spread across $n_{\text{step}} = 2000$ time steps of size $\Delta t = 0.01\tau$, as in Problem 4, with `collide_gases`. Measure the equilibrium temperature of both gases and compare to the prediction of Equation 5: with gravity, gas A has heat capacity $C_A = 2\,U_0/T_0$, and without it gas B has $C_B = 1\,U_0/T_0$, so both should reach $T = 2\,T_0$. (Make sure you do not use `collide_reservoir` while bringing the two gases into equilibrium.) Report the temperatures as `"P6 TA"` and `"P6 TB"`.

[1] Other combinations work, but I found this to be a reliable and reasonably efficient setting.